MANGLARES VIVOS

PROJECT INITIATIVE

NOVEMBER, 2026



GOBIERNO DE LA REPÚBLICA DE HONDURAS

INSTITUTO NACIONAL DE CONSERVACIÓN FORESTAL

I. GENERAL PROJECT IDENTIFICATION

1. Name	Manglares Vivos (Living Mangroves): Hydrological Restoration, Community Governance and Blue Carbon in the Gulf of Fonseca and the Honduran Caribbean
2. Location	Main area: Gulf of Fonseca (departments of Choluteca and Valle), within the Southern Honduras Wetlands System Ramsar Site (Sistema de Humedales de la Zona Sur de Honduras), with 4 intervention sites in at least 2 of its 7 Habitat/Species Management Areas. Caribbean pilot site: Cuero y Salado Wildlife Refuge (Atlántida), Ramsar Site No. 619, where the project rehabilitates silted channels and protects the habitat of the Antillean manatee. Cuero y Salado remains a pilot within this project because it is the only site in the country with a siltation assessment carried out after Hurricanes Eta and Iota and with dredging permits in process, and because it makes it possible to compare the cost and results of hydrological rehabilitation on both coasts; its conceptual design and environmental license will be prepared during the development of the Project Design Document (PDD). During the PDD preparation, a shortlist of the Gulf sites will be presented, including their area by stratum, land tenure, and mapped shrimp farming and salt production concessions. Since the Gulf of Fonseca is a trinational system, the project will coordinate with the Gulf's trinational bodies (Honduras, El Salvador and Nicaragua) on hydrodynamics, sediments and fisheries (A2.6).
3. Duration	5 years of implementation (2027–2031). Year 1 is devoted to start-up, the baseline and the hydrological assessment. Civil works are carried out during the dry-season window of each coast, defined with data from COPECO and the National Meteorological Service (SMN): March to May 2029 in Cuero y Salado, with a contingency season in March–May 2030; and December to April in 2029–2030 and 2030–2031 in the Gulf of Fonseca, following the design and the environmental license. Year 5 is devoted to consolidating restoration, monitoring and re-measuring the carbon plots.
4. Estimated total cost	**HNL 176,718,120.58, equivalent to US$ 6,538,577.00** Grant. Exchange rate HNL 27.0270 per US$ (Central Bank of Honduras, BCH, 2 Oct 2026).
Implementing entity	National Institute for Forest Conservation, Protected Areas and Wildlife (ICF), through its Project Coordination Office.

Suggested implementation channel	Technical implementation by ICF through its Pacific and Atlantic regional offices and its Department of Protected Areas, together with the protected area co-managers (CODDEFFAGOLF in the southern zone and FUCSA in Cuero y Salado) and national academic institutions: the Pacific Coast Regional University Center (CURLP) and the Atlantic Coast Regional University Center (CURLA) of the National Autonomous University of Honduras (UNAH), and the National University of Forest Sciences (UNACIFOR). Funds administered by the development partner or by an accredited entity; the fiduciary arrangement will be defined with the development partner in the PDD. Civil works are contracted by ICF through public tender, with independent supervision contracted separately. Component 3 will be coordinated with the National Directorate of Climate Change (DNCC) of the Ministry of Natural Resources and Environment (SERNA), which is responsible for the national greenhouse gas (GHG) inventory. Governance: a Steering Committee (ICF, development partner, SERNA/DNCC, co-managers and academia) approves the annual operating plans (AOPs) and reviews the risk register quarterly, and a technical committee for each site coordinates field implementation. The ICF–co-manager and ICF–academia agreements, with terms of reference and deliverables, and the ICF–SERNA/DNCC letter of understanding will be conditions precedent to the first disbursement. The PDD will include the organizational chart, the RACI matrix by activity and the Project Charter.
National counterpart contribution	In kind, estimated at HNL 11,445,000 (≈ US$ 423,465) over the 5 years: ICF technical and coordination staff at the central level, in the forest regions and in the local offices, as well as institutional facilities and equipment. Indicative amount, subject to confirmation by the Project Coordination Office (OCP) and to valuation based on payroll records.
Potential partners	The Pew Charitable Trusts and The Nature Conservancy (TNC), within the framework of the Mangrove Breakthrough; Germany's International Climate Initiative (IKI), with GIZ as a possible implementing agency; the Global Mangrove Alliance; and MAR Fund for the Cuero y Salado pilot site. At the national level: SERNA/DNCC, the Directorate General of Fisheries and Aquaculture of the Ministry of Agriculture and Livestock (SAG-DIGEPESCA), the National Aquaculture Association of Honduras (ANDAH), CODDEFFAGOLF and FUCSA.

Contribution to NDC 3.0	Expected net removals of 2.4–3.3 Gg CO_2e in biomass in 2031 (4.1–5.6 Gg CO_2e including soil carbon), based on the establishment curve of the works, and a potential of 3.0–4.1 Gg CO_2e per year once established (5.2–7.0 Gg including soil; 0.6–1.4% of the target), calculated by climate zone and verified through permanent plots; no carbon credits generated$_2e$ including soil; 0.6–1.4% of the LULUCF target of 503.8 Gg CO_2e by 2030). Calculations are stratified by climate zone (dry Pacific and humid Caribbean) using the values of the IPCC 2013 Wetlands Supplement, and methane emissions from low-salinity sites are deducted; results are verified through permanent plots. Results are accounted for in the national GHG inventory, whose Forest Reference Emission Level (FREL) already includes mangroves; the project does not generate or sell carbon credits, which avoids double counting. It initiates restoration processes on 800 ha of mangroves, which count toward the indicator of NDC 3.0 Adaptation Measure 12, and generates national data on carbon in mangrove biomass and soil.
Contribution to watershed management	Protects the lower reaches of the Choluteca, Nacaome and Goascorán river basins, which flow into the Gulf of Fonseca: mangroves retain sediments and regulate the exchange between fresh and salt water. Sedimentation from the upper watersheds is addressed in the design of the works and in coordination with watershed management projects such as FOREST-TEC.
Contribution to protected areas	Implemented in the Southern Honduras Wetlands System Ramsar Site and in the Cuero y Salado Wildlife Refuge; its results, including mangrove migration corridors, are incorporated into the management plans of these areas. In Cuero y Salado, it protects the habitat of the Antillean manatee, a conservation target of the refuge.
Integration with SIGMOF	Restored areas, blue carbon plots, water level and salinity sensor data, and surveillance and complaint records are entered into SIGMOF.
Contribution to adaptation	Implements the NDC 3.0 adaptation measures to restore at least 10% of mangroves in viable areas and to maintain or improve the ecological integrity of wetlands in conservation areas, and measures an adaptation outcome: the catch of mangrove cockles (curiles) at restored sites compared with control sites. It responds to the exposure of the Pacific coast to sea-level rise, saline intrusion, storm surges and El Niño-related droughts, as identified in the First Adaptation Communication (2025), and secures corridors for the landward migration of mangroves.
Strategic alignment	Articles 5 and 7 of the Paris Agreement; IPCC 2013 Wetlands Supplement; NDC 3.0 restoration target and adaptation measures on mangroves, wetlands and marine-coastal areas; First Adaptation Communication (2025); National Adaptation Plan (2018); National Climate Change Strategy (2010); National Wetlands Policy 2018–2038; National Biodiversity Policy 2018–2030; National Drought Risk Reduction Plan 2020–2038; National Forest Restoration Strategy 2023–2030; Ramsar Convention. Management Plan of the Southern Zone Protected Areas Subsystem 2015–2026 (ICF Agreement 028-2015) and its update. ICF Sustainable Tourism Strategy (in the process of official adoption).

Contribution to NDC 3.0

NDC 3.0 target or measure	Project contribution	Project indicator
LULUCF target: reduce net emissions by 1% by 2030 (503.8 Gg CO_2e), through restoration that recovers the water regime of degraded areas	Restoration of 800 ha of mangroves: expected net removals of 2.4–3.3 Gg CO_2e in biomass in 2031 and a potential of 3.0–4.1 Gg CO_2e per year once established (5.2–7.0 Gg including soil; 0.6–1.4% of the target), calculated by climate zone and verified through permanent plots; no carbon credits generated	I.2 Net CO_2 removals in restored areas
Adaptation Measure 12: restore at least 10% of mangroves in viable areas by 2035, using the Manual for Assessing Natural Regeneration in Mangrove Forests	800 ha with restoration initiated, located in the viable areas identified by the project assessment (A1.2) and evaluated using the ICF Manual. The national area of viable zones, which is the basis for this measure's indicator, is defined by ICF at the national level; the project's percentage contribution will be calculated against it.	R.1 Area with restored hydroperiod
Adaptation Measure 13: maintain or improve the ecological integrity of at least 65% of conservation areas with wetlands by 2035	Restoration, surveillance and regulated public use in the intervention areas; baseline and monitoring of ecological integrity using the SINAPH guide, and of management effectiveness using the official ICF manual	I.3 Management effectiveness; I.4 Ecological integrity
Adaptation Measure 18 (tourism): develop at least five tourism development plans with a climate change approach by 2035	Complementary contribution: the regulation of public use under the ICF Sustainable Tourism Strategy provides inputs for these plans	PR.2.1 Areas applying the Sustainable Tourism Strategy

Targets and measures as per the official text of NDC 3.0 (2026); measure numbering follows the technical annex. Indicators correspond to the Results Matrix of this profile.

Logical Framework Matrix

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
Goal: To contribute to the NDC 3.0 LULUCF target, restoration target and adaptation measures on mangroves, wetlands and marine-coastal areas.	**F.1** National mangrove carbon data accepted by the national GHG inventory team (SERNA/DNCC) in 2030 and updated with plot re-measurement in 2031, for use at IPCC Tier 2. **F.2** Expected net removals of 2.4–3.3 Gg CO_2e in biomass in 2031 in the areas restored up to 2030, based on the establishment curve, verified with SIGMOF and project plots; potential of 3.0–4.1 Gg CO_2e per year in biomass once the 800 ha are established (complementary data). **F.3** Ecological integrity of mangroves at intervention sites equal to or better than the baseline in 2031 (SINAPH Ecological Integrity Monitoring Guide; NDC 3.0 Adaptation Measure 13); the management effectiveness of protected areas (ICF Agreement 037-A-2013) is reported as a contribution indicator, with no attributable target.	National GHG inventory; SIGMOF; project plots; ICF ecological integrity and management effectiveness reports; BTR 2032 (subsequent verification).	National policy maintains the priority given to mangroves in NDC 3.0.

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Purpose: The prioritized mangroves of the Gulf of Fonseca and Cuero y Salado recover their hydrological and ecological functionality under community governance and provide adaptation services to coastal communities.	**P.1** 800 ha with the target hydroperiod restored in 2031 (flooding frequency and duration and salinity range defined for each site relative to a reference mangrove, in line with SER standards), verified with sensors, and regeneration density by cohort meeting the differentiated target according to the year of intervention. **P.2** 300 families with supplementary income from employment in restoration, surveillance or community-based tourism in 2031, with at least 40% of daily wages paid to women. **P.3** 100% of the properties within the hydraulic footprint of the design with an agreement signed and formalized before the works, and ≥ 70% of other adjacent properties with an agreement in force in 2031 (base number defined in the baseline). **P.4** Catch per unit effort of mangrove cockles (curiles) at restored sites higher than at control sites in 2031 (BACI design), as an adaptation indicator.	Monitoring reports; plots and satellite imagery; daily wage payrolls; agreement records; community fisheries monitoring; ICF resolutions.	Communities and producers maintain their participation.

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
Component 1: Hydrological restoration of mangroves.	**C1.1** Project baseline, including control sites, published in 2028. **C1.2** 800 ha with the target hydroperiod restored in 2031 (100 ha in Cuero y Salado in 2029 and 700 ha in the Gulf of Fonseca: 350 ha in the 2029–2030 dry-season window and 350 ha in the 2030–2031 window), verified with water level and salinity sensors. **C1.3** 110 ha with direct propagule planting in 2031. **C1.4** Regeneration density by cohort in 2031: ≥ 2,625 plants per ha (ENRF minimum for mangroves) in the 2029 cohort, ≥ [1,800] in the 2030 cohort and ≥ [1,000] in the 2031 cohort. **C1.5** Register of unit costs of hydrological rehabilitation published in 2029 (Cuero y Salado) and updated in 2031 (Gulf of Fonseca). **C1.6** 5 sites with mangrove migration corridors incorporated into management plans or municipal zoning in 2030. **C1.7** ≈ 44,000 workdays paid to local families by 2031, at least 40% to women, under a labor management procedure (minimum age of 18, protective equipment and accident insurance).	Monitoring reports; sensor records; works supervision reports; ICF restoration certificates.	Adjacent landowners and concession holders respect the formalized agreements.

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
Component 2: Local governance, livelihoods and public use.	**C2.1** 100% of properties within the hydraulic footprint with a formalized agreement (notarized or by administrative resolution) before the works at each site, ≥ 70% of other adjacent properties with a signed agreement in 2029, and livelihood restoration plan implemented. **C2.2** ≥ 70% of complaints of illegal impacts receiving an institutional response from ICF or the Environmental Prosecutor's Office, and hectares of detected and addressed impacts reported annually from 2028, with ≥ 2 patrols per month per site. **C2.3** ICF Sustainable Tourism Strategy applied in 3 protected areas in 2029, and 2 community guide groups operating in 2030. **C2.4** Community manatee monitoring protocol operating in Cuero y Salado from 2028, protection protocol during the works and boat speed ordinance processed in 2029; collisions and entanglements recorded annually. **C2.5** 4 trinational technical meetings held in the Gulf of Fonseca by 2031.	Agreement records; surveillance logs; complaint register; ICF resolution; group and sighting records; trinational meeting minutes.	Security and access conditions allow work at the prioritized sites.

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
Component 3: National blue carbon data.	**C3.1** Mangrove sampling intensification protocol adopted in 2029. **C3.2** 60 permanent carbon plots (40 project plots and 20 national plots, with sample size by objective and stratum) installed and measured in 2028, before the works, and re-measured in 2031. **C3.3** Interim data delivery protocol in 2028; biomass and soil carbon data and first national growth rates (using chronosequences) published, and MRV protocol adopted, in 2031.	Technical report; plot database; adoption resolutions.	Inventory institutions maintain their participation under the ICF–SERNA/DNCC letter of understanding.
A1.1 Project baseline: assessment of natural regeneration using the ICF Manual (current density), stratification of mangroves into areas without cover and degraded mangrove (canopy fraction), ecological integrity of the intervention areas, control sites for the BACI design (including mangrove cockle catch) and household socioeconomic baseline, completed before the first works window.	Baseline published · **US$ 150,000.00**	Baseline report; financial and progress reports.	Access to sites during the appropriate season

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
A1.2 Hydrological assessment with RTK topographic survey and drone photogrammetry, tide, water level and salinity sensors, hydraulic model and site-specific designs, climate risk analysis using AR6 scenarios (sea-level rise, hypersalinity, drought and hurricanes), identification of viable areas and mangrove migration corridors and their incorporation into management plans and municipal zoning, and estimation of the re-siltation rate.	Designs and risk analyses for 5 sites; corridors incorporated into management plans or zoning · **US$ 300,000.00**	Technical studies; incorporation records; financial and progress reports.	Hydrological and tidal data available
A1.3 Rehabilitation of silted channels in Cuero y Salado (dredging), designed with manatee habitat criteria, with a dredged material management plan and a unit cost register.	100 ha with restored hydroperiod; unit cost register · **US$ 576,000.00**	Works and supervision reports; cost report; financial and progress reports.	Environmental license and dredging permit approved during preparation
A1.4 Rehabilitation of channels and water flows in the Gulf of Fonseca (desilting and opening of flows using local labor and an amphibious excavator), with self-cleaning design or sediment traps, an excavated material management plan and a unit cost register.	700 ha with restored hydroperiod; unit cost register · **US$ 384,000.00**	Works and supervision reports; cost report; financial and progress reports.	Environmental licenses approved within the expected timeframe
A1.5 Direct propagule planting only where there is no natural regeneration, in the rainy season following hydrological rehabilitation.	110 ha planted with local labor · **US$ 200,000.00**	Planting records and SIGMOF; financial and progress reports.	Availability of local propagules

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
A1.6 Maintenance and monitoring of the hydroperiod, channels and regeneration by cohort, with water level and salinity sensors at each site and an operation and maintenance plan for sensors and channels (without duplicating the planting maintenance under A1.5).	Target hydroperiod verified and sensors operating at 100% of sites · **US$ 400,000.00**	Monitoring reports; sensor records; financial and progress reports.	Community participation in monitoring
A2.1 Implementation of the ICF Sustainable Tourism Strategy in the project's protected areas: regulation of public use (carrying capacity at prioritized sites, visitation and filming rules, signage) and staff training.	Strategy applied in 3 protected areas · **US$ 150,000.00**	Implementation reports; ICF resolution; financial and progress reports.	The ICF Sustainable Tourism Strategy is officially adopted
A2.2 Dialogue and agreements with adjacent shrimp farmers, salt producers and cattle ranchers: legal mapping of land tenure, concessions and uses; dialogue roundtable for each site with ANDAH and artisanal salt producers; agreements formalized before a notary or by administrative resolution (shared channels that improve water exchange in the ponds, no expansion of embankments into the mangrove, and buffer strips), separate from the legal compliance track before ICF, DIGEPESCA and the Environmental Prosecutor's Office; and a livelihood restoration plan for those who lose current uses.	100% of properties in the hydraulic footprint before each work and ≥ 70% of other adjacent properties with an agreement; livelihood restoration plan implemented · **US$ 300,000.00**	Map; agreement records; plan reports; financial and progress reports.	Producers are willing to engage in dialogue

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
A2.3 Community surveillance against logging, landfilling and embankment expansion (daily wages, fuel and equipment), without confrontation, with an anonymous complaint protocol linked to ICF and the Environmental Prosecutor's Office, and a financial sustainability mechanism signed before 2030.	≥ 70% of complaints with institutional response, ≥ 2 patrols per month per site and financial mechanism signed · **US$ 450,000.00**	Surveillance logs; complaint register in SIGMOF; financial and progress reports.	Communities organized for surveillance
A2.4 Pilot community-based tourism, subject to a demand study: guide groups (prioritizing women and youth), training and basic equipment in Cuero y Salado and at a pilot site in the Gulf, with a transitional measure for maintaining the Boca del Toro infrastructure with FUCSA in 2027–2028.	2 guide groups operating; Boca del Toro infrastructure maintained · **US$ 170,000.00**	Demand study; group and visitor records; financial and progress reports.	The demand study confirms feasibility
A2.5 Protection of Antillean manatee habitat in Cuero y Salado: community sighting monitoring with FUCSA, signage, boat speed ordinance with the municipality, manatee protection protocol during the works (observers, exclusion zones and schedules) and entanglement prevention with fishers.	Monitoring protocol operating; ordinance processed; protocol applied during the works · **US$ 120,000.00**	Sighting, collision and entanglement records; minutes; financial and progress reports.	Participation of fishers and boat operators
A2.6 Trinational coordination in the Gulf of Fonseca: annual technical meetings with the Gulf's trinational bodies and exchange of information on hydrodynamics, sediments and fisheries with El Salvador and Nicaragua.	4 trinational technical meetings and data exchange reports · **US$ 50,000.00**	Meeting minutes; reports; financial and progress reports.	Trinational bodies are willing to participate

Narrative summary of objectives	Verifiable indicator	Means of verification	Assumptions
A3.1 Protocol to intensify mangrove sampling, including the islands of the north coast, and a network of 60 permanent carbon plots with academia (40 project plots and 20 national plots), with sample size by objective and stratum, installed and measured before the works.	Protocol adopted and 60 plots installed and measured · **US$ 400,000.00**	Protocol; plot database; financial and progress reports.	Access to sites at suitable tides
A3.2 Laboratory analysis of soil and biomass carbon (including the carbon content of dredged material) and preliminary growth estimates using chronosequences.	Data published · **US$ 250,000.00**	Technical report; financial and progress reports.	National or regional laboratory available
A3.3 Re-measurement of the 60 plots and first national mangrove growth rates.	60 plots re-measured and first rates published · **US$ 130,000.00**	Plot database; technical report; financial and progress reports.	Permanent plots preserved
A3.4 Interim data delivery protocol (2028), MRV protocol and prior legal analysis, with SERNA and the Office of the Attorney General, of blue carbon rights and benefit sharing, aligned with the national REDD+ framework and Article 6.	Interim protocol in 2028 and MRV protocol adopted in 2031 · **US$ 200,000.00**	ICF resolutions; financial and progress reports.	Participation of inventory institutions

Vertical logic: activities produce the components; components achieve the purpose; the purpose contributes to the goal, provided that the assumptions at each level hold. For activities, the indicator is the physical target and its cost.